

JHU Engineering for Professionals
Applied and Computational Mathematics
Data Mining 625.740

Homework for Module 2

1. Let X have an exponential distribution

$$p(x|\vartheta) = \begin{cases} \vartheta e^{-\vartheta x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Sketch $p(x|\vartheta)$ versus x for a fixed value of the parameter ϑ .
- (b) Sketch $p(x|\vartheta)$ versus $\vartheta, \vartheta > 0$ for a fixed value of x .
- (c) Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|\vartheta)$. Show that the maximum likelihood estimate for ϑ is given by

$$\hat{\vartheta} = \frac{1}{\frac{1}{n} \sum_{k=1}^n x_k}.$$

2. Let X have a uniform distribution

$$p(x|\vartheta) = \begin{cases} \frac{1}{\vartheta}, & 0 \leq x \leq \vartheta \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Sketch $p(x|\vartheta)$ versus ϑ for an arbitrary value of x .
 - (b) Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|\vartheta)$. Show that the maximum likelihood estimate for ϑ is $\max_k x_k$.
 - (c) Find the method of moments estimator for ϑ .
3. Let $\mathbf{X}$ be a binary $(0, 1)$ vector with multivariate Bernoulli distribution

$$p(\mathbf{x}|\boldsymbol{\vartheta}) = \prod_{i=1}^d \vartheta_i^{x_i} (1 - \vartheta_i)^{1-x_i},$$

where $\boldsymbol{\vartheta} = (\vartheta_1, \dots, \vartheta_d)^T$ is an unknown parameter vector, ϑ_i being the probability that $x_i = 1$. Show that the maximum likelihood estimate for $\boldsymbol{\vartheta}$ is

$$\hat{\boldsymbol{\vartheta}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k.$$

4. Let X have a gamma distribution ($\Gamma(x; \alpha, \beta)$)

$$p(x|\alpha, \beta) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, & x > 0 \text{ and } \alpha, \beta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Suppose that n samples $x_1, \dots, x_n$ are drawn independently according to $p(x|\alpha, \beta)$. Find the method of moments estimator for α and β .
- (b) Show that the exponential distribution is $\Gamma(x; 1, 1/\vartheta)$.
5. Let the random variables X and Y be independent with exponential distributions

$$p(x|\alpha) = \begin{cases} \alpha e^{-\alpha x}, & x \geq 0 \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad p(y|\beta) = \begin{cases} \beta e^{-\beta y}, & y \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Find $p_z(z|\alpha, \beta)$, the distribution of $Z = X + Y$.

6. Let X have a normal distribution with mean μ and variance σ^2 ($X \sim \mathcal{N}(\mu, \sigma^2)$)

$$p(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}.$$

Let $Y = e^X$.

- (a) Find the distribution $p(y|\mu, \sigma)$ of Y .
- (b) What is the mean of Y ?
- (c) What is the variance of Y ?